Ex:No: 4A Date: 18·03·26	Predicting Crop Disease Risk Using Decision Tree Algorithm

Problem Description:

An agricultural research institute monitors environmental conditions to reduce crop losses caused by plant diseases. Factors such as temperature, humidity, rainfall, and soil moisture strongly influence disease occurrence. The institute wants to **predict whether crops are at high or low disease risk** based on these conditions using a **Decision Tree model**, enabling early preventive measures.

Objective:

To build and evaluate a Decision Tree classification model in R to predict crop disease risk using environmental data with train–test split and accuracy calculation.

Procedure:

- **Step 1: Create the Crop Environment Dataset - crop_data**

	Temperature	Humidity	Rainfall	SoilMoisture	DiseaseRisk
1	25	80	120	High	High
2	30	60	40	Medium	Low
3	28	75	90	High	High
4	35	55	30	Low	Low
5	22	85	150	High	High
6	27	78	100	High	High
7	33	50	20	Low	Low
8	29	70	85	Medium	High
9	31	65	60	Medium	Low
10	24	88	140	High	High
11	34	52	25	Low	Low
12	26	82	110	High	High

```
crop_data <- data.frame(
Temperature = c(25, 30, 28, 35, 22, 27, 33, 29, 31, 24, 34, 26),
Humidity = c(80, 60, 75, 55, 85, 78, 50, 70, 65, 88, 52, 82),
Rainfall = c(120, 40, 90, 30, 150, 100, 20, 85, 60, 140, 25, 110),
SoilMoisture = c("High", "Medium", "High", "Low", "High",
                 "Low", "Medium", "Medium", "High", "Low",
DiseaseRisk = c("High", "Low", "High", "Low", "High", "High",
                "High", "Low", "High", "Low", "High")
)
```

#Print crop_data

```
print(crop_data)
```

OUTPUT:

	Temperatures	Humidity	Rainfall	Soil Moisture	Disease Ris
1	25	80	120	High	High
2	30	60	40	Medium	Low
3	28	75	90	High	High
4	35	55	30	Low	Low
5	22	85	150	High	High
6	27	78	100	High	High
7	33	50	20	Low	Low
8	29	70	85	Medium	High
9	31	65	60	Medium	Low
10	24	88	140	High	High
11	34	52	25	Low	Low
12	26	82	110	High	High

- **Step 2: Convert Categorical Variables to Factors**

```
crop_data$SoilMoisture <- as.factor(crop_data$SoilMoisture)
crop_data$DiseaseRisk <- as.factor(crop_data$DiseaseRisk)
```

```
# str() - crop_data
str(crop_data)
```

Step 3: Train–Test Split – 70% training and 30% Testing

```
index <- sample(1:nrow(crop_data), 0.7* nrow(crop_data))
train_data <- crop_data[index,]
test_data <- crop_data[-index,]
```

```
#Print number of rows in train and test data
cat("Training rows: ", nrow(train_data), "\n")
cat("Testing rows: ", nrow(test_data), "\n").
```

- **Step 4: Build the Decision Tree Model - disease_tree**

```
library(rpart)
library(rpart.plot)
disease_tree <- rpart(
DiseaseRisk ~ Temperature + Humidity + Rainfall + SoilMoisture,
data = train_data
method = "class",
control = rpart.control(minsplit = 2, cp = 0)
)
```

'data.frame': 12 obs. of 5 variables:

$ Temperature : num 25 30 28 35 22 27 33 29 31 24 .....

$ Humidity : num 80 60 75 55 85 78 50 70 65 88 .....

$ Rainfall : num 120 40 90 30 150 100 20 85 60 140 .....

$ SoilMoisture : Factor w/3 levels "High", "Low", "Medium" :
1 3 1 2 1 1 2 3 3

$ DiseaseRisk : Factor w/2 levels "High", "Low", 1 2 1 2 1 1 2 1 2 1
.....

Training rows: 8

Testing rows: 4

#Print disease_tree

```
print (disease_tree)
```

- **Step 5: Visualize the Tree**

```
rpart.plot (disease_tree)
```

- **Step 6: Predict Disease Risk**

```
predicted_risk <- predict (disease_tree, test_data,
                           type = "class")
Predicted_risk
```

- **Step 7: Confusion Matrix**

```
confusion_matrix <- table (
    Actual = test_data $ DiseaseRisk,
    Predicted = predicted_risk
)
confusion_matrix
```

- **Step 8: Accuracy Calculation**

```
accuracy <- sum(diag(confusion_matrix))/
            sum (confusion_matrix)
accuracy.
```

Result:

Thus, the program successfully demonstrates the use of the **Decision Tree algorithm in R** to predict crop disease risk based on environmental conditions. The model effectively classifies disease risk using a **train–test split and accuracy evaluation**, providing valuable insights for early agricultural intervention.

n = 8

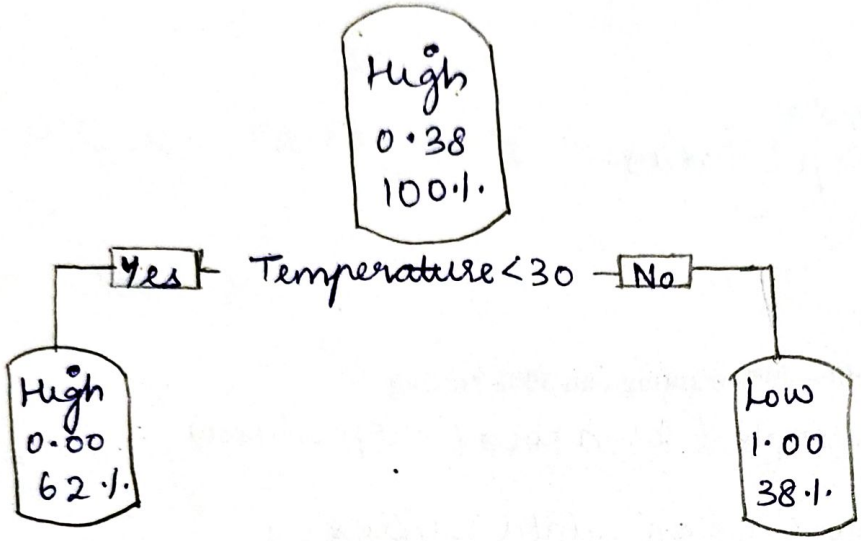
node), split, n, loss, yval, (yprob)

∴ denotes terminal node

1) root 8 2 High (0.7500000 0.2500000)

2) Temperature < 30 6 0 High (1.0000000 0.0000000) ∴

3) Temperature >= 30 2 0 Low (0.0000000 1.0000000) ∴



> predicted_risk

2	4	5	7
Low	Low	High	Low

Levels: High Low

> accuracy

[1] 1

> confusion_matrix

	Predicted	
Actual	High	Low
High	1	0
Low	0	3